Quality Assurance vs Testing

Key Takeaways:

Quality Assurance	Testing
Proactive: defect prevention	Reactive: defect detection
Process-oriented:	Product-oriented

> QA is a process-oriented approach that includes defining, improving, and ensuring adherence to best practices, standards, and procedures throughout the software development lifecycle. QA involves activities such as process audits, quality planning, documentation reviews, training, and continuous improvement initiatives. Instead of directly evaluating the final product, QA focuses on refining the development processes to ensure they are efficient and effective, resulting in more stable and dependable software. Quality Assurance vs Testing | BrowserStack	> Software Testing is a product-focused activity aimed at evaluating a software to identify defects, ensure functionality, and verify that it meets specified requirements. It involves executing the software under controlled conditions to detect errors, bugs, or unexpected behavior before it reaches end users. Testing can be manual or automated and spans various levels, such as unit testing, integration testing, system testing, and acceptance testing. The core goal of Software Testing is to validate the quality of the final product and confirm that it behaves as expected in real-world scenarios. Quality Assurance vs Testing | BrowserStack
• Quality Assurance is **a continuous process** that begins	• Software testing is the process of executing a program or system

during the planning stages of a product or service and continues through the development, testing, and deployment of the product or service	with the intent of finding defects or errors. Testing requires that a piece if software already be written/developed

- Differences between Software Testing and Quality Assurance - GeeksforGeeks
  - “Software testing and quality assurance (QA) are two related but distinct processes within the software development life cycle.“
  - “Quality Assurance, on the other hand, is a broader term focusing on the overall process of ensuring that a product or service meets its design specifications and meets the user’s needs. Quality Assurance is a continuous process that begins during the planning stages of a product or service and continues through the development, testing, and deployment of the product or service.“
  - “Quality assurance is focused on the implementation of processes, procedures, and standards to ensure the quality of the software.“
  - “Software testing is the process of executing a program or system with the intent of finding defects or errors. It is done to verify the functionality of the software product and to ensure it meets the requirements and design specifications. “
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- Council Post: The Distinction Between Testing And Quality Assurance In The Software Industry
  - “QA is in control of the testing process while it ensures that the software’s performance is effective under certain conditions. Testing’s focus is evaluation, case studies and implementation. QA is short for quality assurance, and it is common knowledge that product quality cannot be “tested.” Therefore, quality assurance merely optimizes the process so that quality can be achieved. Manual tests and their respective reports are often used to assess the success of such quality measures.“
  - “QA is a management approach that takes into account all the policies, processes, standards, training, tools and so on that are necessary to ensure the achievement of a company’s quality objectives. … Testing is only a tiny part of this strategy.“
  - “When QA connects with testing, it is referred to as quality control (QC). … QC is a collaborative process, one in which teams work together from inception and at every stage

of software development, focusing on quality. Quality means everyone taking responsibility for their work from the second production begins to the second it ends.“

- Quality Assurance in Software Development - CodeSuite
  - key factors in quality assurance in software development to maximize quality:
    - **Define Quality Goals and Standards**
    - **Develop test plans and strategies**
- Complete guide to quality assurance in software development
  - Quality assurance (QA) is a **process** that ensures a company’s products meet **set quality standards.** QA activities involve meeting **specific industry requirements**, as well as meeting the **needs and expectations of customers and end-users**.
  - There are a few different **quality assurance models** you can follow to ensure quality assurance throughout the entire software development cycle
  - “Put simply, QA is process-oriented and focused on preventing defects, and QC is product-oriented and focused on identifying defects.“